MIL-I-23413C(SH)
4 August 1981
SUPERSEDING
MIL-I-23413B(SHIPS)
2 February 1968
(See 6.5)

MILITARY SPECIFICATION

INSERTS, WELDING, FILLER MATERIAL, COILED AND SOLID RINGS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers consumable insert material in coil and solid ring forms designed for replacement in weld joints fabricated from carbon steel, alloy steels, aluminum alloy, and nickel base alloys (including copper nickel alloy (70/30)).

1.2 Classification.

1.2.1 Classes and styles. Inserts shall be furnished in the following classes and styles, as specified (see 6.2):

Class 1 - Inverted T-shaped cross section (see figure 1).

Style A - Coiled insert filler material.
Style B - Preformed rings, open lap-joint.
Style C - Preformed rings, open butt-joint.

Class 2 - J-shaped cross section (see figure 1).

Style A - Coiled insert filler material.
Style B - Preformed rings, open lap-joint.
Style C - Preformed rings, open butt-joint.

Class 3 - Rectangular-shaped cross section (see figure 2).

Style D - Solid ring, 3/16-inch rim width.
Style E - Solid ring, 1/8-inch rim width.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3439

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Class 4 - Y-shaped cross section (see figure 3).

Style A - Coiled insert filler material.
Style C - Preformed rings, open butt-joint.

Class 5 - Cross section shall be as specified (see 6.2).

Style A - Coiled insert filler material.
Style B - Preformed rings, open-lap-joint.
Style C - Preformed rings, open butt-joint.
Style D - Solid rings.

1.2.2 Types and sizes.

1.2.2.1 Types. Inserts shall be furnished in the types shown in table I, as specified (see 6.2).

TABLE I. Types.

Class	Insert types	Type designation identification	Alloy group identification
All	MIL-Ms-1	Ms-1	Group A (mild steel)
	MIL-Ms-2	Ms-2	
All	MIL-515 (1-1/4 Cr-1/2 Mo)	515	Group B (chromium molybdenum steel)
	MIL-521 (2-1/4 Cr-1 Mo)	521	
	MIL-505 (5 Cr-1/2 Mo)	505	
All	MIL-308	308	Group C (austenitic chromium nickel steels)
	MIL-308Co	308Co	
	MIL-308L	308L	
	MIL-308CoL	308CoL	
	MIL-310	310	
	MIL-312	312	
	MIL-312Co	312Co	
	MIL-316	316	
	MIL-316L	316L	
	MIL-348	348	
	MIL-348Co	348Co	
All	MIL-1100	1100	Group D (aluminum alloys)
	MIL-4043	4043	
	MIL-5356	5356	
All	MIL-60	60	Group E (nickel alloys)
	MIL-61	61	
	MIL-62	62	
	MIL-67	67	
	MIL-82	82	
	MIL-6A	6A	

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1.2.2.2 Sizes. Inserts shall be furnished in the following sizes, as specified (see 6.2):

Class and style	Figure	Size reference
Class 1 and 2, style A	1	3.2.2
Class 1 and 2, style B	1	3.2.3, table VIII
Class 1 and 2, style C	1	3.2.4
Class 3, style D	2	3.3.2, table IX
Class 3, style E	2	3.3.2, table X
Class 4, style A	3	3.4.2
Class 4, style C	3	3.4.3
Class 5, style A	-	3.5.2
Class 5, style B	-	3.5.3, table VIII
Class 5, style C	-	3.5.4
Class 5, style D	-	3.5.5

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-766 - Steel Plates, Sheets, and Strip - Corrosion Resisting.

MILITARY

MIL-W-10430 - Welding Rods and Electrodes; Preparation for Delivery of.

MIL-I-45208 - Inspection System Requirements.

STANDARDS

FEDERAL

FED-STD-151 - Metals; Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-271 - Nondestructive Testing Requirements for Metals.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip.

A 240 - Heat-Resisting Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Fusion-Welded Unfired Pressure Vessels.

(Application for copies should be addressed to the American Society For Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Ratings, Rules, and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 General requirements.

3.1.1 Materials. Insert materials shall be of uniform quality and condition and shall be free of piping, seams, segregations, and abrasions which would adversely affect the usability characteristics of the insert or properties of the fused metal.

3.1.1.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.1.2 Finish. Inserts shall have smoothly finished surfaces and shall be free from slivers, depressions, scratches, and scale or any foreign matter that would adversely affect the usability characteristics of the insert or properties of the fused metal.

3.1.2.1 Rust preventatives. Where it is required, rust preventatives may be used as protective coating provided that they are readily removable with solvents by wiping or dipping, and provided that when a rust preventative has been used, cleaning procedures for removing the rust preventative are clearly indicated on the packaging.

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3.1.2.2 Special requirements for group C and group E inserts.

3.1.2.2.1 Group C and group E inserts shall not be grit or vapor blasted.

3.1.2.2.2 When specified (see 6.2), group C and group E inserts shall be bright annealed. When specified (see 6.2), MIL-62 and MIL-82 inserts shall be special bright hydrogen annealed at 1800°F ± 50°F in dry hydrogen having an entrance dew point of minus 80°F or lower. Following bright annealing or special bright hydrogen annealing, inserts shall not be subjected to any surface treatment other than solvent cleaning.

3.1.3 Chemical composition. The chemical compositions of inserts shall be as specified in tables II through VI.

TABLE II. Chemical composition - mild steel. 1/

	Type	Carbon	Manganese	Phos-phorus	Sulfur	Silicon	Aluminum 2/	Zirconium 2/	Titanium 2/
		Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent
Group A	MIL-Ms-1	0.06	1.00-1.40	0.025	0.03	0.40-0.60	0.05-0.15	0.02-0.12	0.05-0.15
	MIL-Ms-2	.20	0.90-1.30	.025	.03	.45- .65	----	-----	-----

1/ Single values are maximum percentages, except where otherwise specified.
2/ Al + Zr + Ti = 0.15 minimum.

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TABLE III. Chemical composition - chromium molybdenum steels. 1/

	Type	Carbon	Manganese	Phosphorus	Sulfur	Silicon	Aluminum	Chromium	Molybdenum	Copper	Nickel
		Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent
Group B	MIL-515	0.10	0.40-0.70	0.03	0.03	0.40-0.70	0.15	1.2-1.5	0.40-0.65	0.15	
	MIL-521	.10	.40-0.70	.03	.03	.25-0.50	.15	2.3-2.7	.90-1.1	.15	0.40
	MIL-505	.10	.40-0.75	.025	.025	.25-0.50	.15	5.0-6.0	.45-0.65		

1/ Single values are maximum percentages, except where otherwise specified.

TABLE IV. Chemical composition - austenitic chromium nickel steels. 1/

	Type	Carbon	Chromium	Nickel	Molybdenum	Columbium plus tantalum	Manganese	Silicon	Phos-phorus	Sul-fur	Co-balt	Chromium to nickel ratio (minimum)
		Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent
	2/ MIL-308	0.08	19.5-22.0	9.0-11.0	----	----	1.0-2.5	0.25-0.60	0.03	0.03	----	1.9/1
	2/ MIL-308Co	.08	19.5-22.0	9.0-11.0	----	----	1.0-2.5	.25-0.60	.03	.03	0.10	1.9/1
Group C	2/ MIL-308L	.03	19.5-22.0	9.0-11.0	----	----	1.0-2.5	.25-0.60	.03	.03	----	1.9/1
	2/ MIL-308CoL	.03	19.5-22.0	9.0-11.0	----	----	1.0-2.5	.25-0.60	.03	.03	0.10	1.9/1
	MIL-310	.08-0.15	25.0-28.0	20.0-22.5	----	----	1.0-2.5	.25-0.60	.03	.03	----	-----
	MIL-312	.08-0.15	28.0-32.0	8.0-10.5	----	----	1.0-2.5	.25-0.60	.03	.03	----	-----
	MIL-312Co	.08-0.15	28.0-32.0	8.0-10.5	----	----	1.0-2.5	.25-0.60	.03	.03	0.10	-----
	MIL-316	.08	18.0-20.0	11.0-14.0	2.0-3.0	----	1.0-2.5	.25-0.60	.03	.03	----	-----
	MIL-316L	.03	18.0-20.0	11.0-14.0	2.0-3.0	----	1.0-2.5	.25-0.60	.03	.03	----	-----
	2/ MIL-348	.08	19.0-21.5	9.0-11.0	----	10 x C Min. 3/ to 1.0 Max.	1.0-2.5	.25-0.60	.03	.03	----	1.9/1
	2/ MIL-348Co	.08	19.0-21.5	9.0-11.0	----	10 x C Min. 3/ to 1.0 Max.	1.0-2.5	.25-0.60	.03	.03	0.10	1.9/1

1/ Single values shown are maximum percentages, except where otherwise specified.
2/ Delta ferrite content of weld metal (fused insert material) shall be from 4 to 9 percent determined by the Magne-Gage, or equivalent (see figure 5 and notes).
3/ Tantalum content shall not exceed 0.10 percent.

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TABLE V. Chemical composition - aluminum alloys. [1/]

	Type [2/]	Copper	Magnesium	Manganese	Chromium	Silicon	Iron	Zinc	Titanium	Berylium	Other		Aluminum
											Each	Total	
		Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent
Group D	MIL-1100	0.02	----	0.05	----	[3/]	[3/]	0.10	----	0.0008	0.05	0.15	99.0-99.5
	MIL-4043	.30	0.05	.05	----	4.5-6.0	0.8	.10	0.20	.0008	.05	.15	Remainder
	MIL-5356	.10	4.5-5.5	.05-0.20	.05-0.20	[4/]	[4/]	.10	.06-0.20	.0008	.05	.15	Remainder

1/ Single values shown are maximum percentages, except where otherwise specified.
2/ The alloy designations are in accordance with the system approved by the Aluminum Association.
3/ Iron plus silicon = 1.0 percent maximum.
4/ Iron plus silicon = 0.50 percent maximum.

TABLE VI. Chemical composition - nickel alloys (includes copper-nickel). [1/2/]

	Type	Carbon [3/]	Manganese	Iron	Sulfur	Phos-phorus	Silicon	Copper	Nickel plus cobalt	Aluminum	Titanium	Chromium	Columbium plus tantalum	Other elements total
		Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent
Group E	MIL-60	0.15	4.00	2.50	0.02	0.020	1.50	Balance	62.0-69.0	1.25	1.5-3.0	----	----	0.50
	MIL-61	.15	1.00	1.00	.01	.030	0.75	0.25	93.0 minimum	1.50	2.0-3.5	----	----	.50
	MIL-62	.08	1.00	6.0-8.0	.015	.030	.35	.50	70.0 minimum [4/]	----	----	14.0-17.0	[5/] 1.5-3.0	1.0
	MIL-67	.04	1.00	0.4-0.7	.01	.020	.15	Balance	29.0-32.0	----	0.2-0.5	----	----	0.50
	MIL-6A	.08	2.00-2.75	8.00	.015	.030	.35	.50	67.0 minimum [4/]	----	2.5-3.5	14.0-17.0	----	.50
	MIL-82	.10	2.50-3.50	3.00	.015	.030	.50	.50	67.0 minimum [4/]	----	0.75	18.0-22.0	[5/] 2.0-3.0	.50

1/ Analysis shall be made for the elements for which specific values are shown. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine that the total of these other elements is not present in excess of the limits specified for "Other elements, total".
2/ Single values shown are maximum percentages, except where otherwise specified.
3/ When specified (see 6.2), a minimum carbon content shall be required.
4/ Cobalt shall not exceed 0.10 percent.
5/ Tantalum shall not exceed 0.30 percent.

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3.1.4 Alloy identity. Each end of wire to be spliced during processing shall be tested for alloy identity just prior to splicing at the process station (see 4.4.4), except when splicing is done to repair a wire break without removing the wire from the process line.

3.1.4.1 Both ends of each coil drawn to finish size shall be tested for alloy identity (see 4.4.4) before rewinding, spooling, or forming into solid ring inserts.

3.1.4.1.1 For continuous process operation where rod coil is drawn to finish size, straightened and cut to length without removal from the machine, both ends of each rod coil shall be alloy tested immediately prior to the start of the continuous processing operation.

3.1.4.1.2 When rod coils are to be spliced during continuous processing operations, each end of each rod coil to be spliced shall be alloy identity tested at the process station just prior to splicing. In addition, the leading end of the first coil and the tail end of the last coil for each continuous process run shall be alloy identity tested. When alloy identity testing is accomplished in accordance with 3.1.4.1.1 these requirements are satisfied as well.

3.1.4.2 When specified (see 6.2), each cut length of wire in preformed insert rings and each end of each spool or coil of insert wire shall be tested for alloy identity in accordance with 4.4.4 after final marking. Identification marking of each ring, spool, or coil tested after final marking and meeting the acceptance criteria of 4.4.4 shall include the words "ALLOY TESTED".

3.2 Requirements - classes 1 and 2.

3.2.1 Shapes. Shape or configurations of cross sections shall be as shown on figure 1.

3.2.1.1 Dimensions and tolerances. The dimensions and tolerances of the cross sections shall be as specified in tables VII and VIII.

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TABLE VII. Classes 1, 2 and 4 inserts - cross sectional dimensions and tolerances.

Insert size	Legend (figures 1 and 3)	1/ Class 1 Inverted T-shaped		1/ Class 2 J-shaped		2/ Class 4 Y-shaped	
		Dimensions	Tolerances	Dimensions	Tolerances	Dimensions	Tolerances
1/8 inch	D	0.125	±0.004	0.086	+0.011 -.005	0.165	±0.010
	W	.047	+.002 -.012	.047	+.002 -.012	0.078	±0.010
	H	.055	+.012 -.002	.055	+.012 -.002	0.140	±0.010
	H/2	---	----	----	----	0.072	±0.010
5/32 inch	D	0.156	±0.005	.110	+0.012 -.010	0.205	±0.015
	W	.063	+.003 -.014	.063	+.003 -.014	0.093	±0.015
	H	.063	+.014 -.003	.063	+.014 -.010	0.175	±0.010
	H/2	----	----	----	----	0.093	±0.010

1/ The offset between the center of the rib (W) and center of the land (D) shall not exceed 0.005 inch.

2/ Additional dimensions and tolerances - class 4 Y-shaped inserts (see figure 3):

		1/8 inch size	5/32 inch size
(a)	Radius of rib	0.044 ± 0.005	0.050 ± 0.005
(b)	Angle between side of rib and center line	1 degree to 2 degrees	1 degree to 2 degrees
(c)	Rib offset	±0.010	±0.015
(d)	Height of rib along center line	0.100 ± 0.010	0.115 ± 0.010
(e)	Angle between top surface of inclined arm and center line	50 degrees ± 5 degrees	50 degrees ± 5 degrees
(f)	Length of inclined arm	0.085 ± 0.010	0.125 ± 0.010

From a practical inspection standpoint, the D, W, H, and H/2 dimensions have the most to do with usability and weldability. The (a) through (f) dimensions are furnished to complete the description of the cross sectional configuration.

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TABLE VIII. Classes 1 and 2, style B - minimum overlap.

Nominal pipe sizes	Circumferential overlap
Inches	Inch
1/2	1/4
3/4	1/4
1	1/4
1-1/4	1/4
1-1/2	1/4
2	1/4
2-1/2	1/4
3	3/8
3-1/2	3/8
4	3/8
5	1/2
6	1/2
8	5/8
10	3/4
12	3/4
14	3/4

3.2.2 Style A - coil details. Unless otherwise specified (see 6.2), coiled inserts shall be furnished in 50 feet lengths wound on a 12-inch rim. The cast of the insert shall be 9-inches nominal diameter.

3.2.3 Style B - dimensions. Preformed rings with open lap-joints shall be made to order and fitted to a template which shall be furnished by the contractor to suit his machining of the pipe inside diameter (i.d.)(see 6.2).

3.2.4 Style C - dimensions. Preformed rings with open butt-joints shall be made to order and fitted to a template which shall be furnished by the contractor to suit his machining of the pipe i.d. (see 6.2).

3.2.5 Identification. The type designation identification (see table I) shall be imprinted or stamped at 6-inch intervals on style A (coiled) inserts, and shall appear at least once on each style B and style C preformed insert at the location shown on figure 1. Imprints shall be legible, resist effacement by contact incidental to normal handling, and shall not interfere with insert usability. The depth of the imprint shall not exceed 0.015 inch.

3.3 Requirements - class 3.

3.3.1 Design. Unless otherwise specified (see 6.2), solid insert rings shall be of the design shown on figure 2.

3.3.2 Dimensions and tolerances. Solid rings shall be furnished in accordance with the dimensions (styles) shown on figure 2 and the i.d. and outside diameter (o.d.) dimensions specified in tables IX and X.

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TABLE IX. Class 3, style D inserts - Dimensions.

Pipe dimensions			Ring diameter[2/]	
Nominal diameter	Schedule number[1/] (see 6.2)	I.d.[2/]	Ring o.d. for nominal pipe diameter	Ring i.d. for nominal pipe diameter
Inches		Inches	Inches	Inches
2	10S	2.157	2.43	2.06
	40	2.067	2.34	1.97
	80	1.939	2.22	1.85
2-1/2	10S	2.635	2.91	2.54
	40	2.469	2.75	2.38
	80	2.323	2.60	2.23
3	10S	3.260	3.54	3.17
	40	3.068	3.35	2.98
	80	2.900	3.18	2.81
3-1/2	10S	3.760	4.04	3.67
	40	3.548	3.82	3.45
	80	3.364	3.64	3.27
4	10S	4.260	4.54	4.17
	40	4.026	4.30	3.93
	80	3.826	4.10	3.73
5	5S	5.345	5.62	5.25
	10S	5.295	5.57	5.20
	40	5.047	5.32	4.95
	80	4.813	5.09	4.72
6	5S	6.407	6.68	6.31
	10S	6.357	6.63	6.26
	40	6.065	6.34	5.97
	80	5.761	6.04	5.67
8	5S	8.407	8.68	8.31
	10S	8.329	8.61	8.24
	40	7.981	8.26	7.89
	80	7.625	7.90	7.53
10	5S	10.482	10.76	10.39
	10S	10.420	10.70	10.33
	40	10.020	10.30	9.93
	80S	9.750	10.03	9.66
	80	9.564	9.84	9.47
12	5S	12.420	12.70	12.33
	10S	12.390	12.67	12.30
	40S	12.000	12.28	11.91
	40	11.938	12.22	11.85
	80S	11.750	12.03	11.66
	80	11.376	11.65	11.28
14	10	13.500	13.78	13.41
	40	13.126	13.40	13.03
	80	12.500	12.78	12.41
16	10	15.500	15.78	15.41
	40	15.000	15.28	14.91
	80	14.314	14.59	14.22

See footnotes at top of next page.

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1/ Schedule number followed by "S" applies to group C types.
2/ Class 3, style D insert rings are normally furnished to match nominal internal diameters of the pipe. Special ring sizes (non-standard) shall be obtainable when specified (see 6.2). The dimensions of such rings are determined as follows:
Ring i.d. = Pipe i.d. - 0.094 inch.
Ring o.d. = Ring i.d. + 0.375 inch.
Where special applications demand rings having 1/8 inch instead of the 3/16 inch dimension shown on figure 2, the narrower width (style E) shall be made available as shown in table X. To determine the required i.d. and o.d. dimensions for the narrower type, substitute 0.250 inch for 0.375 inch in the above equations.

TABLE X. Class 3, style E inserts - Dimensions.

Pipe dimensions			Ring diameters 2/	
Nominal diameter	Schedule number 1/ (see 6.2)	I.d. 2/	Ring o.d. for nominal pipe diameter	Ring i.d. for nominal pipe diameter
Inches		Inches	Inches	Inches
1/4	10S	0.410	0.57	0.32
	40	.364	.52	.27
	80	.302	.46	.21
3/8	10S	.545	.70	.45
	40	.493	.65	.40
	80	.423	.58	.33
1/2	5S	.710	.87	.62
	10S	.674	.83	.58
	40	.622	.78	.53
	80	.546	.70	.45
3/4	5S	.920	1.08	.83
	10S	.884	1.04	.79
	40	.824	0.98	.73
	80	.742	.90	.65
1	5S	1.185	1.34	1.09
	10S	1.097	1.25	1.00
	40	1.049	1.21	0.96
	80	0.957	1.11	.86
1-1/4	5S	1.530	1.69	1.44
	10S	1.442	1.60	1.35
	40	1.380	1.54	1.29
	80	1.278	1.43	1.18
1-1/2	5S	1.770	1.93	1.68
	10S	1.682	1.84	1.59
	40	1.610	1.77	1.52
	80	1.500	1.66	1.41

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TABLE X. Class 3, style E inserts - Dimensions. - Continued

Pipe dimensions			Ring diameters[2/]	
Nominal diameter	Schedule number[1/] (see 6.2)	I.d.[2/]	Ring o.d. for nominal pipe diameter	Ring i.d. for nominal pipe diameter
Inches		Inches	Inches	Inches
2	5S	2.245	2.40	2.15
	10S	2.157	2.31	2.06
	40	2.067	2.22	1.97
	80	1.939	2.10	1.85
2-1/2	5S	2.709	2.87	2.62
3	5S	3.334	3.49	3.24
3-1/2	5S	3.834	3.99	3.74
4	5S	4.334	4.49	4.24

See footnotes at end of table IX.

3.3.3 Flatness. Solid insert rings shall be flat to within 0.015-inch at any location when measured on a surface plate using feeler gages.

3.3.4 Identification. The type designation identification (see table I) and applicable pipe schedule number (see table IX and style D or E) shall be imprinted or stamped on each ring (see figure 2). Imprints shall be legible, resist effacement during normal handling, and shall not interfere with usability of insert. The depth of the imprint shall not exceed 0.015 inch.

3.4 Requirements - class 4.

3.4.1 Shapes. Shape or configuration of cross section shall be as shown on figure 3.

3.4.1.1 Dimensions and tolerances. The dimensions and tolerances of the cross sections shall be as specified in table VII.

3.4.2 Style A - coil details. Unless otherwise specified (see 6.2), coiled inserts shall be furnished in 50-feet lengths wound on a 12-inch rim. The cast of the insert shall be 9-inches nominal diameter.

3.4.3 Style C - dimensions. Preformed rings with open butt-joints will be made to order and fitted to a template which will be furnished by the contracting activity (see 6.2).

3.4.4 Identification. The type designation number shall be imprinted or stamped at 6-inch intervals on style A (coiled) inserts and shall appear at least once on each style C preformed insert which additionally shall bear applicable nominal pipe size. Imprints shall be legible, resist effacement by contact incidental to normal handling, and

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shall not interfere with insert usability. Stamped designation shall have a depth equal to or greater than a Rockwell C hardness impression on the material being marked. Note: It is recommended that as an alternate identification for all inserts, but especially those excessively hardened from finish rolling or those too small for satisfactory imprinting, a "flag-tag" type marker be used for each circular unit of all styles.

3.5 Requirements - class 5.

3.5.1 Shapes. Shape or configuration of cross section shall be as specified (see 6.2).

3.5.1.1 Dimensions and tolerances. The dimensions and tolerances of the cross sections shall be as specified (see 6.2).

3.5.2 Style A - coil details. Unless otherwise specified (see 6.2), coiled inserts shall be furnished in 50-foot lengths wound on a 12-inch rim. The cast of the insert shall be 9-inches nominal diameter.

3.5.3 Style B - dimensions. Preformed rings with open lap-joints shall be made to order and fitted to a template which shall be furnished by the contractor (see 6.2). Circumferential overlap shall be as specified in table VIII. For nominal sizes greater than 14 inches diameter, circumferential overlap shall be as specified (see 6.2).

3.5.4 Style C - dimensions. Preformed rings with open butt-joints shall be made to order and fitted to a template which shall be furnished by the contractor (see 6.2).

3.5.5 Style D. The i.d. and o.d. dimensions and tolerances shall be as specified (see 6.2).

3.5.5.1 Flatness. Solid insert rings shall be flat to within 0.015 inch at all locations when measured on a surface plate using feeler gages.

3.5.6 Identification. The type designation identification (see table I) shall be imprinted or stamped at 6-inch intervals on style A (coiled) inserts and shall appear at least once on each style B and style C preformed insert. The type designation identification (see table I) and pipe schedule number (if applicable, (see tables IX and X)) shall be imprinted or stamped on each style D (solid ring) insert. Imprints shall be legible, resist effacement during normal handling, and shall not interfere with usability of insert. The depth of the imprint shall not exceed 0.015 inch.

3.6 Fused insert contour. The contour of the backside surface of fused insert shall comply with the following requirements:

(a) Reinforcement shall not exceed 3/32 inch.
(b) Concavity shall not exceed 1/32 inch.
(c) The contour shall have a uniform radius and shall blend smoothly into the base metal.

3.6.1 The fused insert shall be free of linear indications as revealed by liquid penetrant inspection.

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3.6.2 Internal weld surface of fused insert showing oxide scale accompanied by a wrinkled or crystalline surface appearance shall be cause for rejection. However, tightly adhering, irridescent temper films shall be considered acceptable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208.

4.1.2 The identification of melts, heats, or lots; the general requirements for test specimens; the rejection of non-conforming lots or specimens; and the retest and resubmittal of the non-conforming lots shall be in accordance with FED-STD-151.

4.1.3 Certification of quality conformance. Certification of quality conformance shall be furnished with each lot of material offered for acceptance. The certification shall include actual data of specified chemical, mechanical, and metallographic tests. Qualitative results of nondestructive tests shall be recorded in the certification. The minimum certification test data shall be as listed in figure 6. The certification of quality conformance may be submitted in the format shown in figure 6.

4.2 Quality conformance inspection. Inserts shall be examined and tested as specified in table XI. Sampling shall be as specified in 4.2.1 through 4.2.1.4.

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TABLE XI. Summary of quality conformance tests.

	Sampling	Test	Requirements
Visual and dimensional examination	4.2.1.3	4.3	3.2 through 3.5
Chemical analyses	4.2.1.4	4.4.1	3.1.3 and tables II through VI
Fusion test	4.2.1.4	4.4.2	3.6 and 3.6.1
Delta ferrite weld test (for some group C inserts)	4.2.1.4	4.4.3	Table IV
Alloy identity test	3.1.4	4.4.4.1	3.1.4

4.2.1 Sampling for quality conformance inspection.

4.2.1.1 Lot. For the purpose of selecting samples for quality conformance inspection, a lot of coiled or solid inserts shall be as defined for each test as specified in 4.2.1.2 through 4.2.1.4.2.

4.2.1.2 Sampling and inspection of containers. Unit packages and shipping containers shall be sampled and inspected in accordance with MIL-W-10430.

4.2.1.3 Sampling for visual and dimensional examination of inserts.

4.2.1.3.1 Classes 1, 2, 4, and 5 - style A. A lot shall be the number of coils having the same cross section (see 1.2.1) produced from the same heat of metal and offered for delivery at the same time. Sample coils shall be selected from each lot in accordance with MIL-STD-105 at inspection level III and shall be examined as specified in 4.3. The acceptable quality level (AQL) shall be 1.5 percent defective.

4.2.1.3.2 Classes 1 and 2, styles B and C, all class 3 and class 4, style C, and class 5, styles B, C and D. A lot shall be the number of preformed ring inserts having the same cross section (see 1.2.1) produced from the same heat of metal and offered for delivery at the same time. Sample inserts shall be selected from each lot in accordance with MIL-STD-105 at inspection level I. The sample inserts shall be selected in equal number from not less than 5 unit containers and examined as specified in 4.3. The AQL shall be 1.5 percent defective.

4.2.1.4 Sampling for chemical analysis, fusion test, and delta ferrite weld test (all classes). Sample specimens shall be selected in accordance with MIL-STD-105 at inspection level S-2 for chemical analysis

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as specified in 4.4.1. Each specimen shall be taken from a different unit container. If any specimen fails to conform for each and every element, the entire lot shall not be offered for delivery.

4.2.1.4.1 Sufficient quantity of all classes of insert material shall be selected for testing in accordance with 4.4.2 and 4.4.3. For testing class 3 inserts, material of appropriate dimensions, representing the heat of metal from which the inserts are to be made may be removed from the base metal stock and used for the test specified in 4.4.2, provided lot control numbers are established for basic stock material and production records are maintained. Such records shall be made available to the Government inspector.

4.2.1.4.2 At the discretion of the Government inspector, chemical analysis and fusion test may be waived on a shipment made from the same heat as a previous shipment which was chemically analyzed and tested and determined to comply with specification requirements.

4.3 Visual and dimensional examination. Sample inserts selected as specified in 4.2.1.3.1 and 4.2.1.3.2 shall be examined and measured to verify conformance to all of the requirements of this specification which do not involve tests. Each coil selected in accordance with 4.2.1.3.1 shall be examined and measured at nine points distributed uniformly throughout its length.

4.4 Test procedures.

4.4.1 Chemical analyses. Samples selected as specified in 4.2.1.4 shall be analyzed by wet chemcial methods in accordance with FED-STD-151, test method 111.2 or spectro chemical methods in accordance with FED-STD-151, test method 112.2.

4.4.2 Fused insert welds. Fused insert welds shall be made as specified on figure 4 and notes applying thereto.

4.4.3 Delta ferrite weld test. Delta ferrite weld test shall be conducted in accordance with figure 5 and notes thereto for each group C MIL type required by table IV. The delta ferrite of weld metal (fused insert material) shall be determined by the Magne-Gage, or equivalent, calibrated against specimens having known delta ferrite content.

4.4.4 Alloy identity.

4.4.4.1 Procedure. The alloy identity test method may include chemical analysis, metal sorting devices, other approved methods or a combination of methods. The test method shall be submitted for approval to the Naval Sea Systems Command (NAVSEA).

4.4.4.2 Acceptance criteria. If the test demonstrates that the material is not of the type specified, the material shall be rejected.

4.5 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

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5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Domestic shipment and early use.

5.1.1 Packaging. Packaging shall be sufficient to afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the using activity and until early use and may conform to the contractor's commercial practice when such meets these requirements.

5.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rates and will afford protection against physical and mechanical damage during direct shipment from the supply source to the using activity for early use. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation and may conform to the contractor's commercial practice when such meets these requirements.

5.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature; National stock number or manufacturer's part number; specification, specification part number, if different from the manufacturer's part number; class, type, size and control number; contract or purchase order number, contractor's name and destination.

5.2 Use of polystyrene (loose-fill) material.

5.2.1 For domestic shipment and early use. Unless otherwise specified (see 6.2), use of polystyrene (loose-fill) material for domestic shipment and early use for packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labelled as follows:

"CAUTION

Contents cushioned etc., with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber, fiberboard or transparent flexible cellular material."

6. NOTES

6.1 Intended use. Coiled insert material or solid rings are intended for use in making sound, smooth, uniform, crack-free root welds where accessibility is limited to one side. Insert alloy composition and the composition of subsequently deposited weld metal should be compatible with base material.

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6.1.1 Class 4, style A - coil details. Class 4 inserts tend to twist when rim is coiled. This is a result of an essentially triangular configuration, narrower at inner rim contact point than at outer surface. However, special coiling conditions may be specified by contract or order (see 6.2).

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class, style, type, and size (for all classes) (see 1.2, 3.2, 3.3, 3.4, and 3.6 and tables VIII, IX, and X).
(c) Shape, dimensions, and tolerances of cross sections of class 5 inserts (see 1.2.1, 3.6.1, and 3.6.1.1).
(d) Whether bright annealing of group C and E inserts is required (see 3.1.2.2.2).
(e) Whether special bright hydrogen annealing of MIL-62 and MIL-82 inserts is required (see 3.1.2.2.2).
(f) Whether minimum carbon content for group E inserts is required (see table VI).
(g) Whether alloy identity testing of each cut length of wire in preformed inserts and each end of each spool or coil of insert wire after final marking is required (see 3.1.4.2).
(h) If style A coiled inserts are to be furnished in other than 50-feet lengths wound on a 12-inch rim (see 3.2.2, 3.4.2, and 3.5.2).
(i) Templates must be furnished on all orders for style B of classes 1, 2, and 5 and style C of classes 1, 2, 4, and 5 (see 3.2.3, 3.2.4, 3.4.3, 3.5.3, and 3.5.4).
(j) If class 3 solid insert rings are to be other than as specified on figure 2 (see 3.3.1).
(k) Pipe schedule number for class 3 and, if applicable, class 5, style D (see tables IX and X).
(l) Whether special ring sizes are required (see table IX).
(m) Shape, dimensions, and tolerances of cross section for class 5 (see 3.5.1 and 3.5.1.1).
(n) Circumferential overlap of class 5, style B inserts of greater than 14 inches nominal diameter (see 3.5.3).
(o) I.d. and o.d. dimensions and tolerances for class 5, style D inserts (see 3.5.5).
(p) Use of polystyrene (loose-fill) material (see 5.2).
(q) Special coiling conditions for class 4, style A, if required (see 6.1.1).

6.3 Inspection after delivery. Post delivery inspection of inserts to determine conformity to this specification and for acceptance thereof is the responsibility of the consignee. When receipt inspection testing types MIL-308, MIL-308Co, MIL-308L, MIL-308CoL, MIL-348 and MIL-348Co, the consignee or user activity may consider the following delta ferrite test result limits acceptable:

(a) A maximum of 12 percent delta ferrite.

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(b) An average minimum level of 3 percent delta ferrite with no individual inspection level of less than 2 percent delta ferrite.

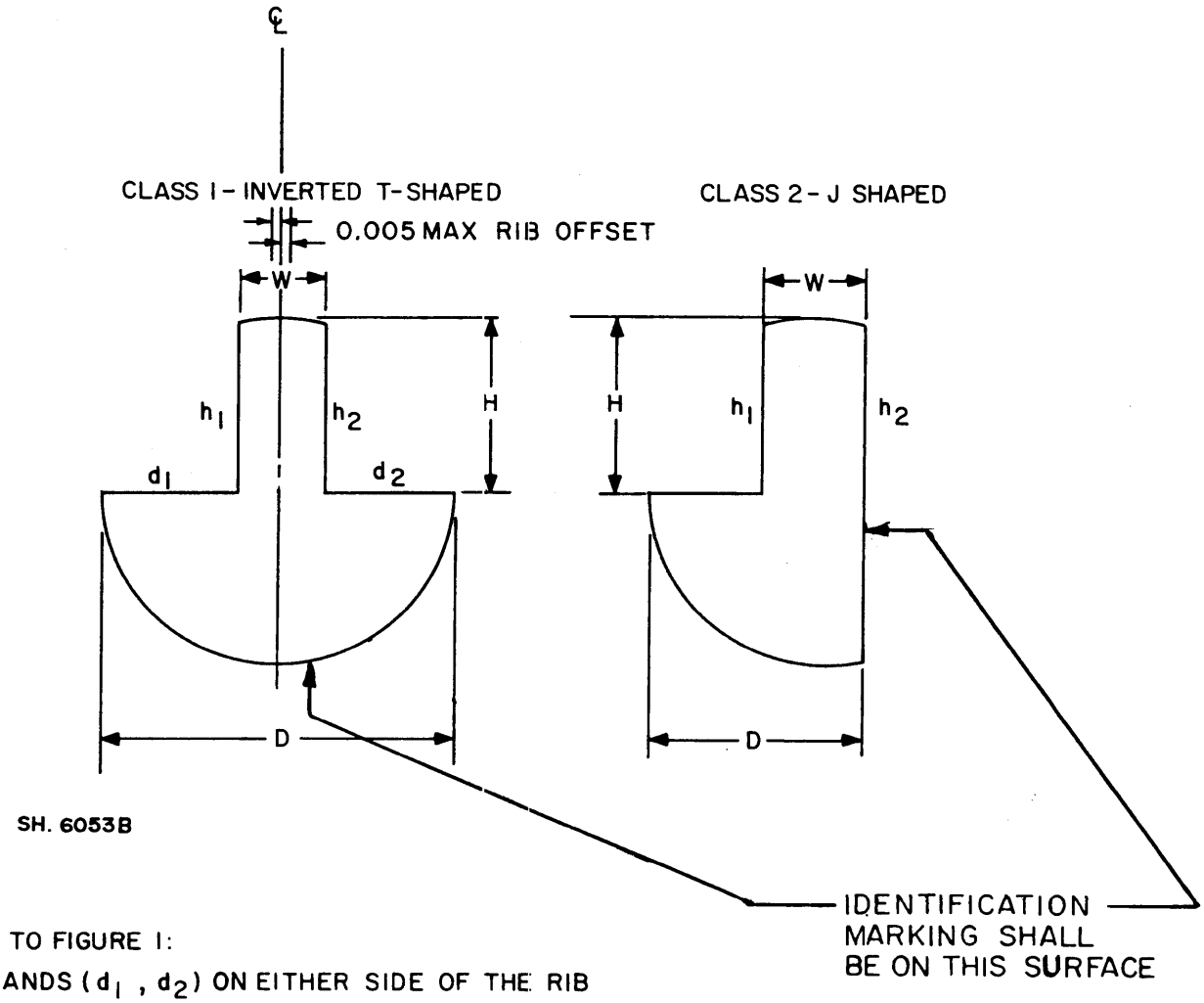
Vendor test results shall meet the specified 4 to 9 percent delta ferrite content.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Changes from previous issue. Asterisks (*) are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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NOTES TO FIGURE 1:

1. LANDS (d_1 , d_2) ON EITHER SIDE OF THE RIB SHALL BE ON THE SAME PLANE WITHIN .005 INCH.

2. RIB SURFACES (h_1 , h_2) SHALL BE PARALLEL WITHIN .002 INCH AND SQUARE WITH LANDS (d_1 , d_2) WITHIN .005 INCH.

FIGURE 1. Class 1 and class 2 inserts - cross sectional configuration.

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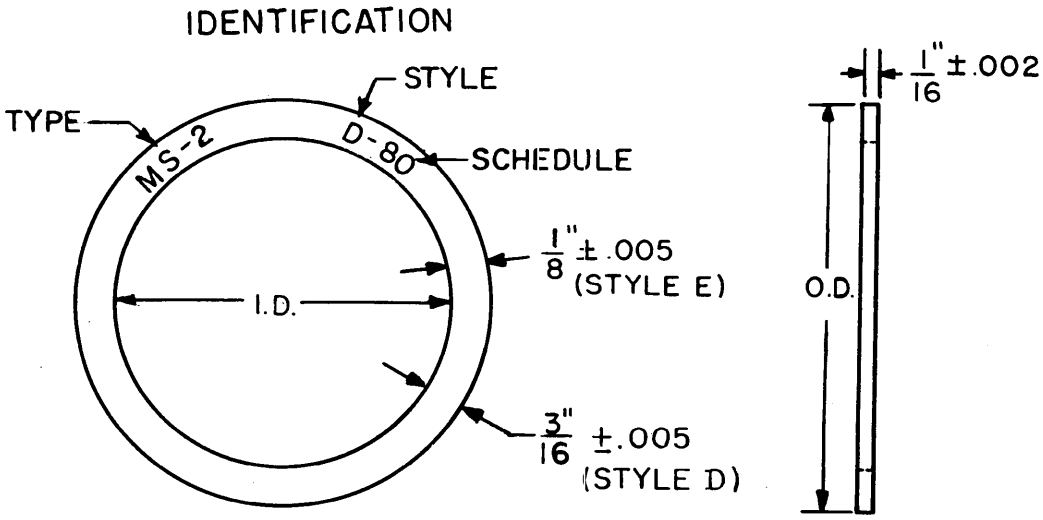


FIGURE 2. Class 3 solid ring inserts.

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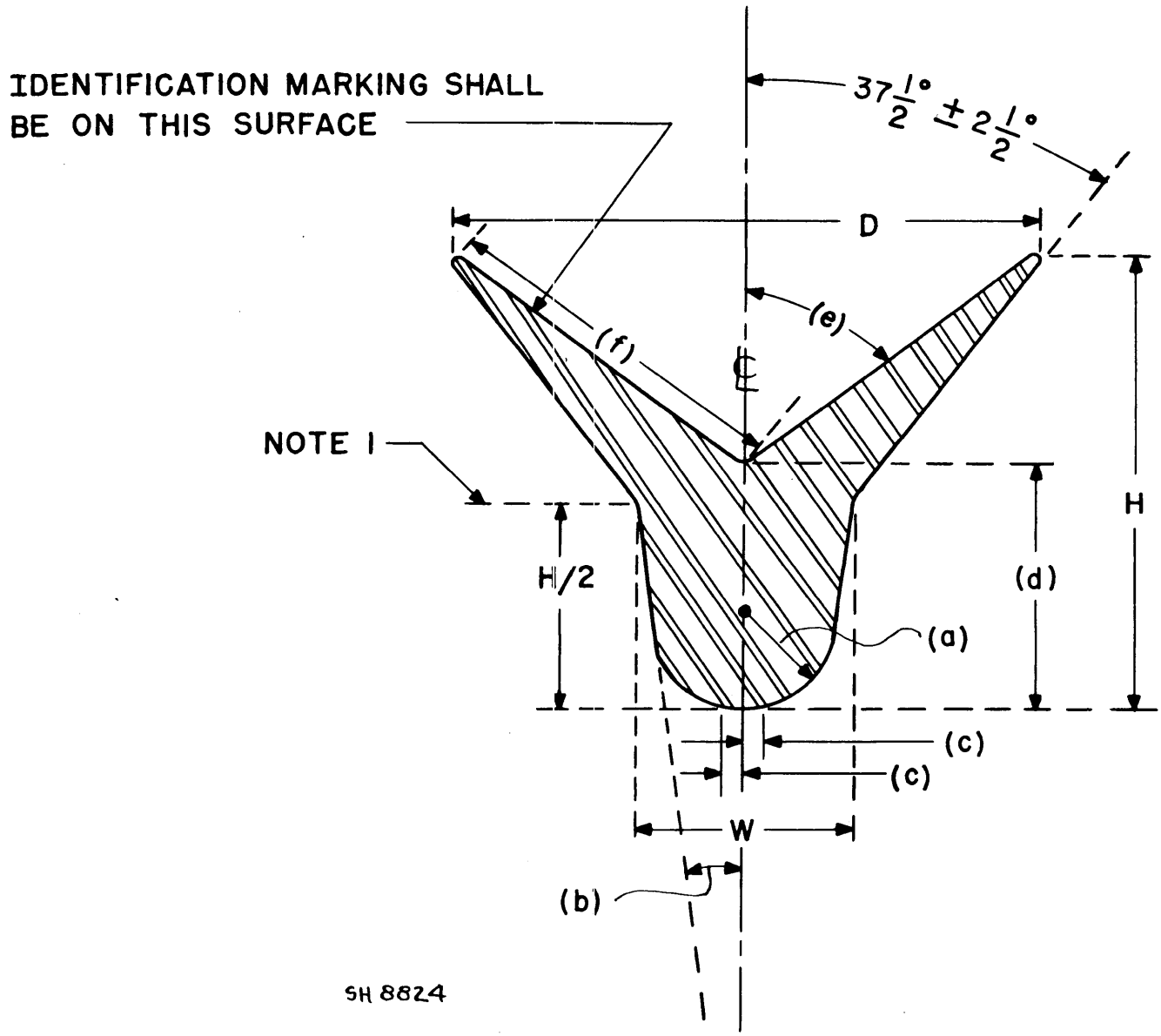


FIGURE 3. Class 4 inserts - cross sectional configuration.

Notes to figure 3:
1. Reference diameter for correlating with pipe i.d.
2. Dimensions and tolerances, see table VII.
3. When specified, rings 1-1/2 inches to 2 inches in diameter shall be formed of 5/32-inch material.

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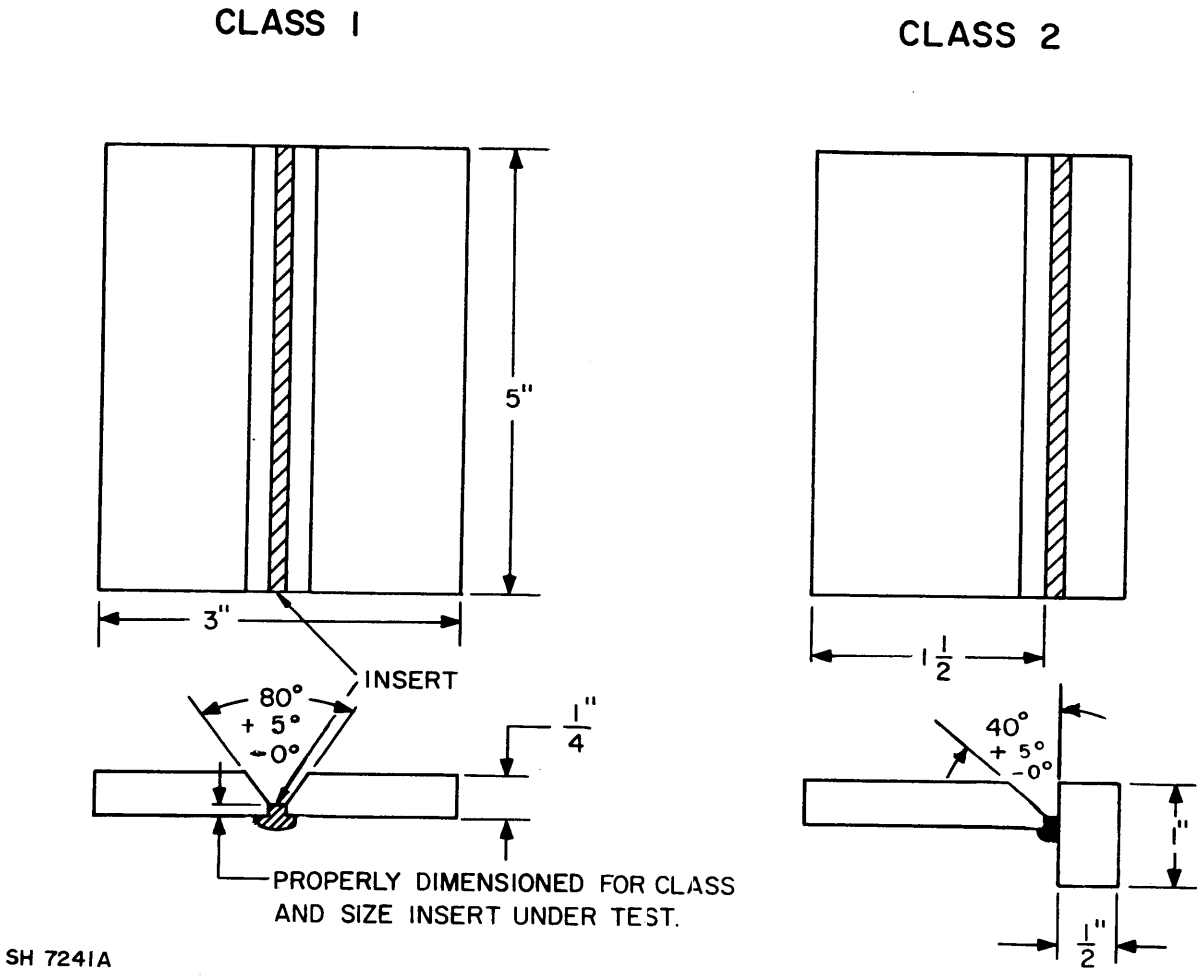


FIGURE 4. Fusibility test joint.

Notes to figure 4:

1. Base plate shall be of composition similar to the insert.
2. One-half of joint shall be welded in flat position, the remainder, in the vertical position. All welding shall be done on groove side.
3. Argon gas shall be used for purging and shielding electrode and back side of weld.
4. Standard techniques and procedures for fusing the insert shall be followed at all times.
5. Upon completion of fusion, the joint shall be visually examined for compliance with 3.6 and shall be liquid penetrant inspected in accordance with MIL-STD-271 with non-water washable penetrant.

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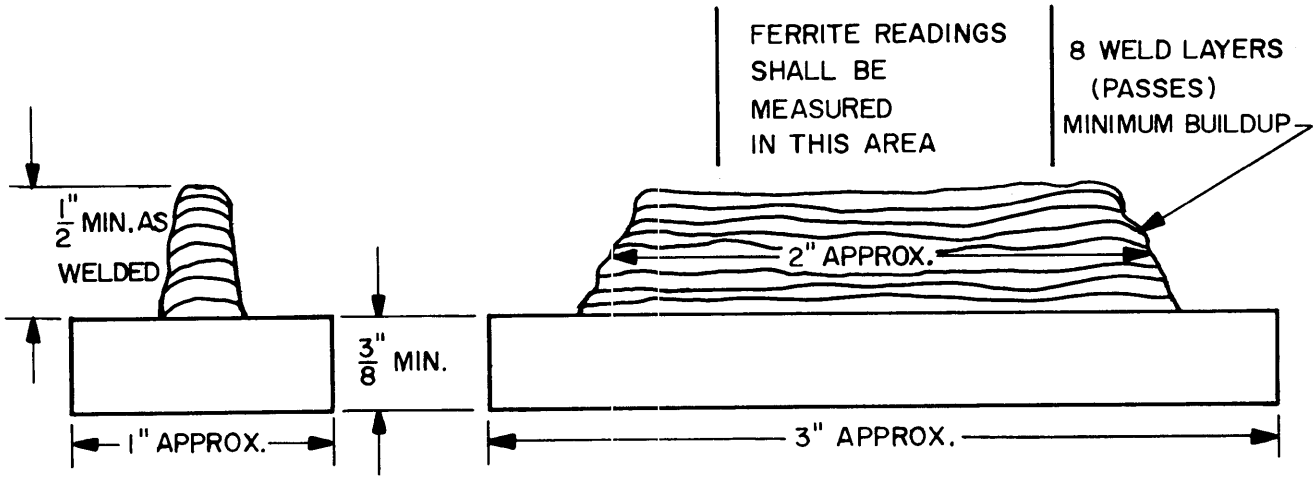


FIGURE 5. Ferrite weld specimen.

Notes to figure 5:

1. The base material for the weld pad shall be types 301, 302, or 304 conforming to QQ-S-766, ASTM A 167, or ASTM A 240.
2. The weld pad shall be built up by depositing single bead layers one on top of the other to a minimum height of 1/2 inch. The weld bead layers shall be deposited with a maximum width in accordance with the table below, except the first several layers may be deposited with a wider weave.
3. The argon gas tungsten-arc welding process shall be used. The weld stops and starts shall be located at the ends of the weld buildup.
4. The weld pad shall be cooled between passes by water quenching immediately after each pass, except the last two weld passes shall be air cooled below 800°F prior to water quenching.
5. Each weld pass shall be cleaned prior to depositing the next weld bead.
6. After welding, the weld buildup shall be surface ground to 64 root mean square (rms) finish to provide sufficient finished surface to make the required ferrite readings. The ground area need not be continuous along the specimen. The weld buildup shall be continuously water cooled during the grinding operation. The finished ground surface shall be approximately parallel with the plate surface and shall be located within the last two passes or layers of weld buildup. Machining and metallographic polishing sufficient to remove the cold working produced by machining may be used in lieu of surface grinding. The weld buildup should be water cooled during the machining operation.

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Notes to figure 5: - Continued

7. A total of six ferrite readings shall be taken on the machined surface along the longitudinal axis of the weld bead. The readings should not be taken on either end where the weld craters (stop and start areas) are located. Reading should be taken in areas having dimensions at least 1-1/4 times the diameter of the magnet. Ferrite readings shall be taken by the Magne-Gage, or equivalent.
8. The six readings obtained shall be averaged to a single value for conversion to percent ferrite.

Insert			Weld bead
Class	Size	Style	Layer width - inches maximum
1	1/8	-	1/4
1	5/32	-	5/16
2	1/8	-	1/4
2	5/32	-	5/16
3	--	D	1/4
3	--	E	5/16
4	1/8	-	1/4
4	5/32	-	3/8

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Manufacturer or distributor _____ Customer Name _____

Address _____ Customer Order No. _____

Date _____ Specification MIL- _____

Heat no. _____ Lot no. _____

Class and style[1/] _____ Size[2/] _____

Type[3/] MIL- _____ Finish[4/] _____

Chemical analysis (report all applicable elements, as specified in tables II, III, IV, V and VI)

Carbon (%) _____ Aluminum (%) _____ Copper (%) _____

Manganese (%) _____ Zirconium (%) _____ Nickel (%) _____

Phosphorus (%) _____ Titanium (%) _____ Columbium plus tantalum (%) _____

Sulfur (%) _____ Chromium (%) _____ Tantalum (%) _____

Silicon (%) _____ Molybdenum (%) _____ Cobalt (%) _____

Iron (%) _____ Other, specify (%) _____

Visual and dimensional examination _____

Alloy identity test _____

Fusibility test (include results of liquid penetrant inspection) _____

(Use additional sheets as necessary)

Delta ferrite (%) (group C inserts only) _____

We hereby certify that the above material has been tested in accordance with the listed specification and is in conformance with all requirements.

_____
Signature of responsible company official

1/ See 1.2.1.
2/ See 1.2.2.2.
3/ See 1.2.2.1.
4/ See 3.1.2.2.

FIGURE 6. Certification of quality conformance.

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   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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